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# **Portable Coordinate Measuring Arm Joint Design**

Mechatronic Project 478  
Final Report

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# Executive summary

| <b>Title of Project</b>                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Joint Design for a Portable Coordinate Measuring Arm (PCMA)                                                                                                                                                                                                           |
| <b>Objectives</b>                                                                                                                                                                                                                                                     |
| To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that maintains and /or improves the arm's overall accuracy.                                                         |
| <b>What is current practice and what are its limitations?</b>                                                                                                                                                                                                         |
| Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.                                                                                                                      |
| <b>What is new in this project?</b>                                                                                                                                                                                                                                   |
| The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.                                                                                                                                                           |
| <b>If the project is successful, how will it make a difference?</b>                                                                                                                                                                                                   |
| It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.                                                           |
| <b>What are the risks to the project being a success? Why is it expected to be successful?</b>                                                                                                                                                                        |
| Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model. |
| <b>What contributions have/will other students made/make?</b>                                                                                                                                                                                                         |
| The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.                                                                                                                                                       |
| <b>Which aspects of the project will carry on after completion and why?</b>                                                                                                                                                                                           |
| The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.                                           |
| <b>What arrangements have been/will be made to expedite continuation?</b>                                                                                                                                                                                             |

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

# **ECSA Exit Level Outcomes**

This section addresses the following Engineering Council of South Africa (ECSA) exit level outcomes and indicates how each outcome was demonstrated in the project.

## **GA 1 - Problem-Solving**

The project addresses an engineering problem, which is the lack of a joint design for PCMA's that meet accuracy requirements that account for spatial dependency and proper position-holding implementation without the impedance of uninhibited movement.

## **GA 2 - Application Scientific and Engineering Knowledge**

## **GA 3 - Engineering Design**

## **GA 5 - Use of Engineering Tools**

## **GA 6 - Professional and Technical Communication**

## **GA 8 - Individual and Collaborative Teamwork**

## **GA 9 - Independent Learning Ability**

# Acknowledgements

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# List of symbols

## Constants

$$L_0 = 300 \text{ mm}$$

## Variables

|            |                                      |                      |
|------------|--------------------------------------|----------------------|
| $Re_D$     | Reynolds number (diameter) . . . . . | [ ]                  |
| $x$        | Coordinate . . . . .                 | [ m ]                |
| $\ddot{x}$ | Acceleration . . . . .               | [ m/s <sup>2</sup> ] |
| $\theta$   | Rotation angle . . . . .             | [ rad ]              |
| $\tau$     | Moment . . . . .                     | [ N·m ]              |

## Vectors and Tensors

$\vec{v}$  Physical vector, see equation ...

## Subscripts

|     |            |
|-----|------------|
| $a$ | Adiabatic  |
| $a$ | Coordinate |

## Abbreviations

|      |                                   |
|------|-----------------------------------|
| 3D   | three-dimensional                 |
| AI   | artificial intelligence           |
| CAD  | computer-aided design             |
| D-H  | Denavit-Hartenberg                |
| DIP  | design independent parameters     |
| FR   | functional requirements           |
| NSoI | narrower system of interest       |
| PCMA | portable coordinate measuring arm |
| SPAT | single-point articulation test    |
| TPM  | technical performance measure     |
| WSol | wider system of interest          |

# Chapter 1

## Introduction

### 1.1 Background

A portable coordinate measuring arm (PCMA) is a handheld metrology instrument that captures three-dimensional (3D) coordinates directly from a physical object, as shown in 1.1 below, (1).



Figure 1.1: A portable coordinate measurement arm, (1)

PCMA's offer adaptable measurement systems for multiple applications, such as dimensional analysis, computer-aided design (CAD) inspection, reverse engineering, etc., (2). The main limitation of PCMA's is the effect of manual operation, which can introduce positional instability and dynamic structural deformation during measurement, (3). Manual operation can also inhibit calibration procedures as it is required that the PCMA must be held still during the procedure, (4).

As a result, designing an effective joint with a position-holding mechanism would improve the design of a PCMA and a good practice of safety. By designing the joint in this manner, error due to manual operation will be minimised, kinematic stability and accuracy will be improved.

## 1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on a PCMA without jeopardising its accuracy and ease of movement.

The objectives of the project are:

1. To generalise the conventional Denavit-Hartenberg(D-H) model derived by Hall and Schreve by sampling the working volume of a PCMA determining distribution of uncertainty along its length, (5).
2. To design, build, and test the joint design through the single-point articulation test (SPAT), such that it maintains and/or improves the accuracy of the PCMA.
3. To compare the results from SPAT to the D-H model and verify the joint design's effect on the PCMA's accuracy.

## 1.3 Scope

In the process of determining D-H parameters, operation manual and available schematics will be used. In addition, only joint will be designed, built, and tested for data acquisition, through calibration and accuracy testing via SPAT.

## 1.4 Motivation

PCMA's are becoming more popular than tradition coordinate measuring machines (CMM) for in-line<sup>1</sup> and on-site industrial measurement, (1). They have broad applicability across multiple industries such as aerospace, automotive, heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite their growing popularity, the PCMA's revolute joints must satisfy competing demands, including:

- Tight run-out tolerances for spatial accuracy.
- Sufficient holding torque to keep the arm stationary once released.
- Internal cable routing.

This has proved difficult since there are found commercial available joints that meet these demands. In addition, it has been established that angular position errors at each joint propagate through the kinematic chain, which affects end-effector's

---

<sup>1</sup>In the context of PCMA's, "in-line" refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory, (1).

accuracy, proving that joints have a measurable impact on the PCMA's accuracy, (6). This project addresses the gap by deriving a joint specifications from a generalised model based off of an error model that was developed by a Stellenbosch Alumnus, Leon Hall, (5). This will produce a validated joint and testing methodology to support future PCMA development.

## **1.5 Use of AI Tools**

Gemini is an artificial intelligence (AI) tool will be used check grammatical errors and sentence structure as well as LaTeX formatting for the report.

## **1.6 Report Overview**

This section follows the following structure:

- Section 1 introduces the project background, aim and objectives, scope, and motivation for project.
- Section 2 presents a literature review discussing sources relevant to PCMA joint design, position-holding technology, kinematic modeling, error analysis, and PCMA testing.
- Section 3 details the PCMA's requirement analysis.

# **Chapter 2**

## **Literature review**

### **2.1 PCMA Context**

Sources such as Leussink Engineering and Meagher gave context as what PCMA's are and how they are used.

### **2.2 The Denavit-Hartenberg Model**



# Chapter 3

## Requirement Analysis

The objective of the requirement analysis is to establish clear expectations of the PCMA's joint design for the purposes of concept development.

### 3.1 Identification of Stakeholders

The first step in the requirement analysis process is to identify stakeholders from the operational and support life cycle phases of the project. Stakeholders will be classified into the following categories, as defined below:

- Primary stakeholders directly interact with the system while it is in operation.
- Secondary stakeholders contribute to the development and functioning of the system.
- Indirect stakeholders rely on or are affected by the system, but do not interact with the system directly.

The stakeholders relevant to the project are defined in Table 3.1 below.

Table 3.1: Identified stakeholders and their classifications

| Stakeholder ID | Description                            | Classification |
|----------------|----------------------------------------|----------------|
| SID_01         | Supervisor of the project              | Primary        |
| SID_02         | Students researching PCMA development  | Secondary      |
| SID_03         | Companies invested in PCMA development | Indirect       |
| SID_04         | Mechanical workshop staff              | Indirect       |
| SID_05         | Mechatronics laboratory managers       | Indirect       |

## 3.2 System Context

The next step is to define the narrower system of interest (NSoI) and the wider system of interest (WSoI) of the project<sup>1</sup>, (7) for the purposes of this project, the joint was identified as the NSoI and WSoI is the NSoI with a reliable power supply and an artefact with which the joint will be tested. Figure 3.1 illustrates the system context, with the NSoI shown as a "black box", (7).

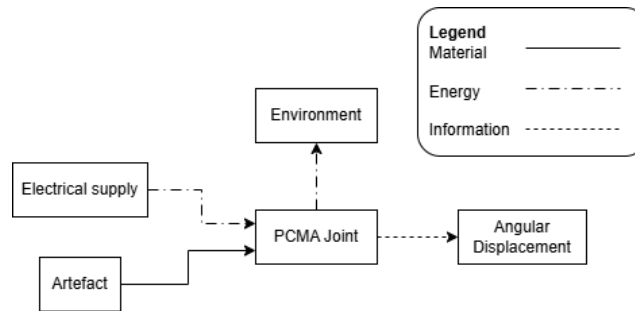


Figure 3.1: System Context

## 3.3 Design Independent Parameters

Next, the design independent parameters (DIPs) are defined. DIPs influence the system during the operational and support life cycle stages, but are not controllable nor are part of the system being defined, (7). The DIPs of the joint design are detailed in Table 3.2 below.

Table 3.2: Design Independent Parameters

| DIP_ID   | Description       | Expected range                               |
|----------|-------------------|----------------------------------------------|
| DIP_01.1 | Electrical supply | 12 V - 24 V                                  |
| DIP_01.2 | Electrical supply | 12 A maximum                                 |
| DIP_01.3 | Electrical supply | Direct current                               |
| DIP_02.1 | Artefact          | Compliant with ISO 10360-12:2016             |
| DIP_02.2 | Artefact          | 20 points reference points                   |
| DIP_03.1 | Environment       | Laboratory ambient temperature: 0 °C - 35 °C |
| DIP_03.2 | Environment       | Laboratory ambient humidity: 20 % - 80 %     |
| DIP_03.3 | Project budget    | R 6000                                       |

<sup>1</sup>The NSoI is the system being designed and the WSoI includes the NSoI and elements that the NSoI interacts with, (7).

### 3.4 Functional Requirements

In this section, the primary and top-level derived functional requirements (FRs) for the joint design are defined. The only FRs that are considered are only associated with joint's operation. FRs are categorised into primary and derived FRs, in which:

- Stakeholders are directly concerned with primary FRs.
- Derived FRs are required so that the system can perform primary FRs.

As seen in Table 3.3 below, the FRs for the joint design are listed. In Figure 3.2,

Table 3.3: Table 3: Functional Requirements

| FR ID | Description                     | Primary / Derived |
|-------|---------------------------------|-------------------|
| FR_01 | Articulate output link          | P                 |
| FR_02 | Hold position                   | P                 |
| FR_03 | Measure angular displacement    | P                 |
| FR_04 | Store angular displacement data | D                 |
| FR_05 | Release position                | P                 |

the functional requirements of the joint are depicted below.

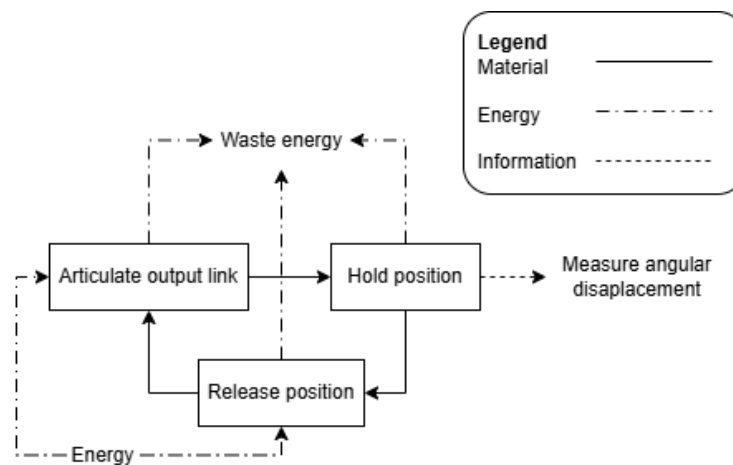


Figure 3.2: Top Level Functional Decomposition

### 3.5 Technical Performance Measures

This section defines the technical performance measures (TPMs) which will be used to determine how well the joint design (NSoI) performs the required

functions<sup>2</sup>, in Table 3.4 below.

Table 3.4: Technical Performance Measures

| TPM ID | Related FR          | Description       | Target range               |
|--------|---------------------|-------------------|----------------------------|
| TPM_01 | FR_01, FR_05        | Run-out tolerance | $1.571 \times 10^{-3}$ rad |
| TPM_02 | FR_02               | Holding torque    | 7.725 Nm                   |
| TPM_03 | FR_03, FR_04        | Encoder accuracy  | 1000 PPR                   |
| TPM_04 | FR_01, FR_02, FR_05 | Joint life cycle  | $10^6$ life cycles         |

---

<sup>2</sup>TPMs are measurable and unambiguous parameters that can be influenced by designing the joint

# **Chapter 4**

## **Concept Development**

### **4.1 Concept Generation**

#### **4.1.1 Concept One:**

#### **4.1.2 Concept Two:**

#### **4.1.3 Concept Three:**

### **4.2 Concept Evaluation**

### **4.3 Concept Selection**

### **4.4 Design Refinement**

# **Chapter 5**

## **Design Analysis**

### **5.1 Mechanical Design**

### **5.2 Electronic Design**

### **5.3 Software Design**

## **Chapter 6**

# **Design Evaluation**

### **6.1 As-Built System Overview**

### **6.2 Performance Testing**

# **Chapter 7**

## **Experimental Procedure**

### **7.1 Experimental Apparatus**

### **7.2 Experimental Setup**

### **7.3 Experimental Equipment Specifications**

#### **7.3.1 Test Sphere Specifications**

#### **7.3.2 Joint Component Specifications**

#### **7.3.3 Experimental Apparatus Specifications**

#### **7.3.4 Methodology**



## **Chapter 8**

## **Conclusions**

# Appendix A

## Mathematical proofs

### A.1 Euler's equation

Euler's equation gives the relationship between the trigonometric functions and the complex exponential function.

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{A.1})$$

Inserting  $\theta = \pi$  in (A.1) results in Euler's identity

$$e^{i\pi} + 1 = 0 \quad (\text{A.2})$$

### A.2 Navier Stokes equation

The Navier–Stokes equations mathematically express momentum balance and conservation of mass for Newtonian fluids. Navier-Stokes equations using tensor notation:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_j} [\rho u_j] = 0 \quad (\text{A.3a})$$

$$\frac{\partial}{\partial t} (\rho u_i) + \frac{\partial}{\partial x_j} [\rho u_i u_j + p \delta_{ij} - \tau_{ji}] = 0, \quad i = 1, 2, 3 \quad (\text{A.3b})$$

$$\frac{\partial}{\partial t} (\rho e_0) + \frac{\partial}{\partial x_j} [\rho u_j e_0 + u_j p + q_j - u_i \tau_{ij}] = 0 \quad (\text{A.3c})$$

## **Appendix B**

### **Experimental results**

# List of references

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